

DAY-9

(8-06-26)

(Q) what is the function $y = A + \bar{B}c$ in PoS form

(a) $M_0 + M_2 + M_3$

(b) $M_0 M_2 M_3$

(c) $M_0 M_1 M_2 M_3$

(d) $M_0 + M_1 + M_2 + M_3$

A. $y = A + \bar{B}c$

~~$(A + \bar{B}c) + (A + \bar{B}c) + A$~~

$= (A + \bar{B}) \cdot (A + c) = (A + \bar{B} + c) \cdot (A + \bar{B} + \bar{c}) \cdot (A + B + c) \cdot (A + \bar{B} + c)$

$= 010, 011, 000, 110$

$= \pi M (0, 2, 3)$

$= M_0 \cdot M_2 \cdot M_3$ [b].

(Q) A 3 Variable table has high o/p for input 010, 011, 110. Boolean exp. can be written as.

A. $010 \rightarrow 1$ (minterms)
 $011 \rightarrow 1$ (SOP)
 $110 \rightarrow 1$

$y = \bar{A}\bar{B}c + \bar{A}Bc + AB\bar{c}$ (SOP)

$= \bar{A}B + AB\bar{c} = B[\bar{A} + A\bar{c}] = \underline{\underline{\bar{A}B + \bar{A}c}}$

(Q) Find o/p y if f_1 & f_2 are

$f_1 = \sum m (2, 3, 4)$ 010 011 100

$f_2 = \pi M (1, 2, 5, 7)$ 001 010 101 111

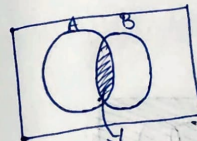
$= \sum m (0, 3, 4, 6)$

$f_1 = \bar{A}\bar{B}c + \bar{A}Bc + A\bar{B}\bar{c}$

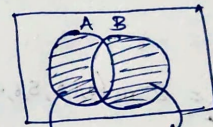
$f_2 = (A + B + \bar{c}) \cdot (A + \bar{B} + c) \cdot (\bar{A} + B + \bar{c}) \cdot (\bar{A} + \bar{B} + \bar{c})$

[Contd...]

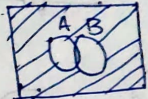
Minterms through logic gates :-



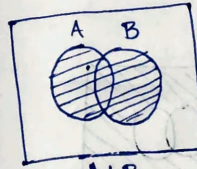
AND



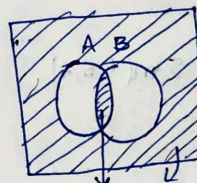
OR



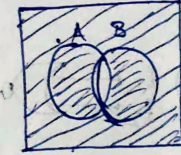
NOR



OR

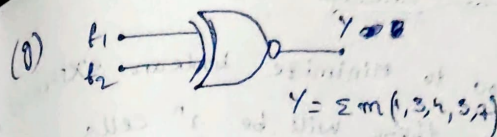
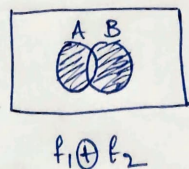
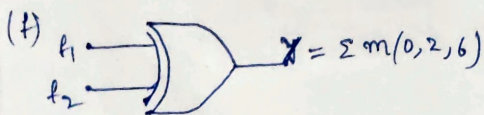
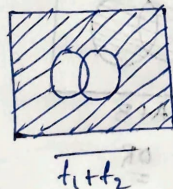
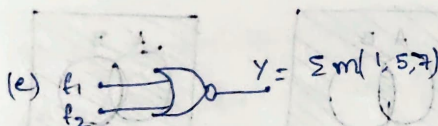
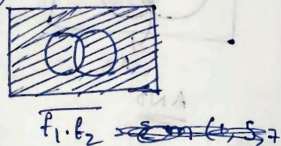
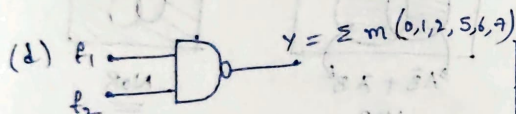
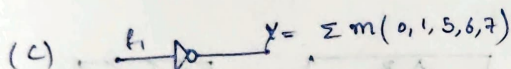
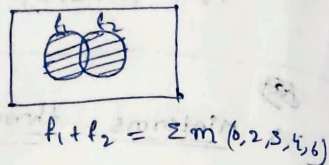
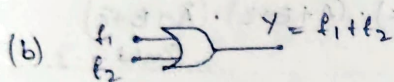
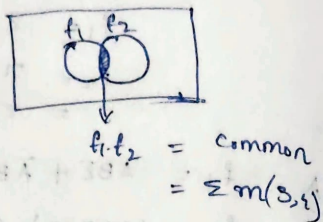
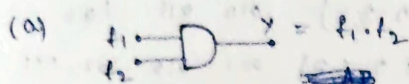


XNOR

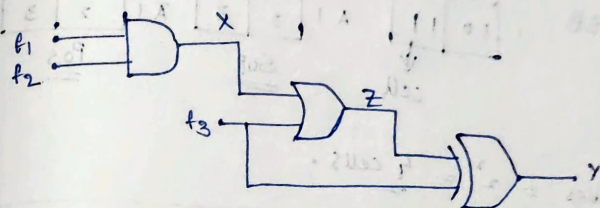


NAND

Continue... Sum



(a) Find y



$$f_1 = \Pi M(2,3,5) = \Sigma m(0,1,4,6,7)$$

$$f_2 = \Pi M(3,6,7) = \Sigma m(0,1,2,4,5)$$

$$f_3 = \Sigma m(1,3,6)$$

$$x = f_1 \cdot f_2 = \Sigma m(0,1,4) \cdot f_1 \cdot f_2$$

$$z = x + f_3 = \Sigma m(0,1,3,4,6)$$

$$y = z \oplus f_3 = \Sigma m(0,4) = \overline{A} \overline{B} \overline{C} + A \overline{B} \overline{C} = \overline{B} \overline{C}$$

K-Maps (Karnaugh Maps)

→ Graphical Method to Minimize boolean Expression.

→ For n Variables, there will be 2^n cells.

(*) 2 Variables:-

A → MSB
B → LSB

B	0	1
A	00	01
1	10	11

cell

SOP

B	0	1
A	0	1
1	2	3

POS

B	0	1
A	0	1
1	2	3

2 Variables = $2^2 = 4$ cells.

(Q) $f(A,B) = \sum m(1,2)$. Fill K-map cells.

A.

B	0	1
A	0	1
1	1	0

(SOP)

3-Variables = $2^3 = 8$ cells (1-bit change)
 $00 \rightarrow 01 \rightarrow 11 \rightarrow 10$

(a)

BC	00	01	11	10
0	000	001	011	010
1	100	101	111	110

A → MSB
C → LSB

(b)

BC	00	01	11	10
0	000	001	011	010
1	100	101	111	110

→

BC	00	01	11	10
0	0	1	3	2
1	4	5	7	6

(c)

ABC	000	001	011	010	100	101	110	111
0	0	1	2	3	4	5	6	7
1	8	9	10	11	12	13	14	15

⇒ determining Value of each cell depends on MSB & LSB, which doesn't change.

(a) $f(A,B,C) = \pi M(2,3,5,7)$. Fill K-Map cells?
 $= \sum m(0,1,4,6)$

A.

BC	00	01	11	10
0	1	1	0	0
1	1	0	0	1

SOP

BC	00	01	11	10
0	1	1		
1	1			1

(Minterms)

POS

BC	00	01	11	10
0			0	0
1		0	0	

(Maxterms)

4- Variables :-

~~ABCD~~

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	8	9	4	10
10	12	13	15	14

(a)

1 → MSB
D → LSB

(Switch these 2 Rows written wrong)

AB \ CD	00	01	11	10
00	0	4	12	8
01	1	5	13	9
11	3	7	15	11
10	2	6	14	10

(b)

Maxterms $\Rightarrow \Sigma M(0, 1, 2, 3, 4, 5, 6, 7, 12, 14)$

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0		0
11	0			0
10	0	0		

(Pos)

(Q) Truth Table is given for $f(A, B, C, D)$.
Fill K-Maps cells.

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

$$Y = \Sigma m(2, 5, 6, 9, 11, 13)$$

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	1	7	6
11	8	1	15	14
10	12	1	1	10

(SoP)

$$Y = \Sigma m(0, 1, 3, 4, 7, 8, 10, 12, 14, 15)$$

(Pos)

AB \ CD	00	01	11	10
00	0	0	0	
01	0		0	
11	0		0	0
10	0			0

(Q) $f(A, B, C, D) = ABED + ABC + ABCD + ABCD$
Fill the K-maps cell?

Minterms $\Rightarrow 1101, 1010, 1011, 1111, 0111$
 $\Rightarrow \Sigma m(10, 11, 13, 15, 7)$

A.

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	1	6
11	8	9	1	10
10	12	13	15	14

(SoP)

(Q) Truth Table for $f(A,B,C,D)$ - Map

A	B	C	D	y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

$y = \sum m(0, 2, 5, 6, 8, 9, 11, 13, 15)$

(SOP)

CD	00	01	11	10
AB 00	1			1
01		1		1
11		1	1	
10	1	1	1	

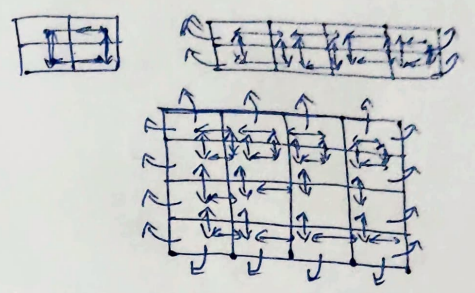
$y = \sum M(1, 3, 4, 7, 10, 12)$

(Pos)

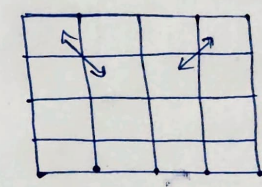
CD	00	01	11	10
AB 00		0	0	
01	0		0	
11	0		0	
10				0

(*) Rules to be followed in K-map :-

(i) club the adjacent cells.

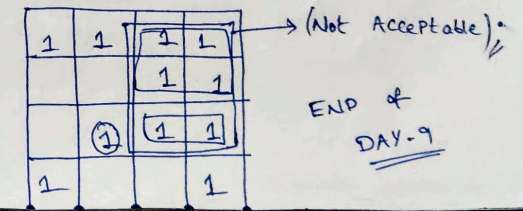


(ii) Don't club diagonal cells.



(*) Between two adjacent cells in K-map, there will be a difference of 1-bit only.

(ii) clubbing can be done in order of $1, 2, 4, \dots, 8, \dots, 2^n$ cells.



END of
DAY-9